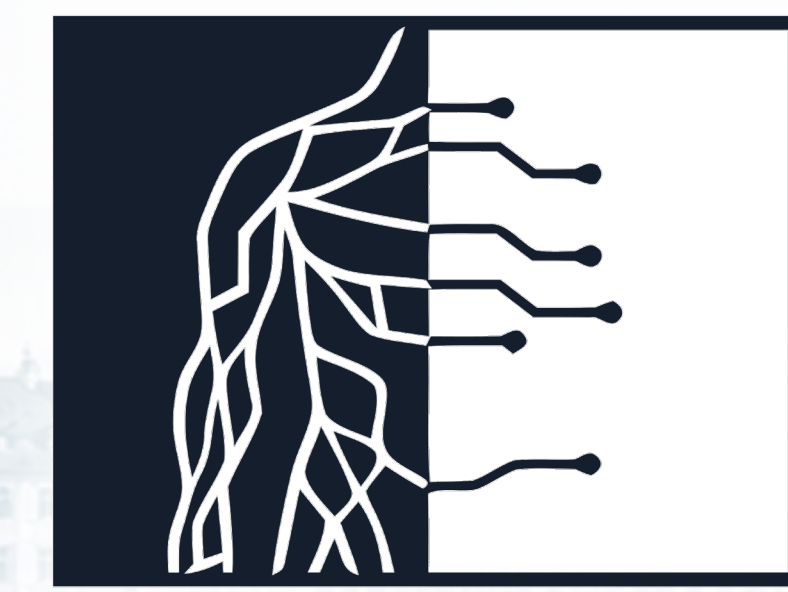


Streak-reduced Human-scale Dark-field CT with 3D Gaussian Splatting



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2026
Bern, Switzerland
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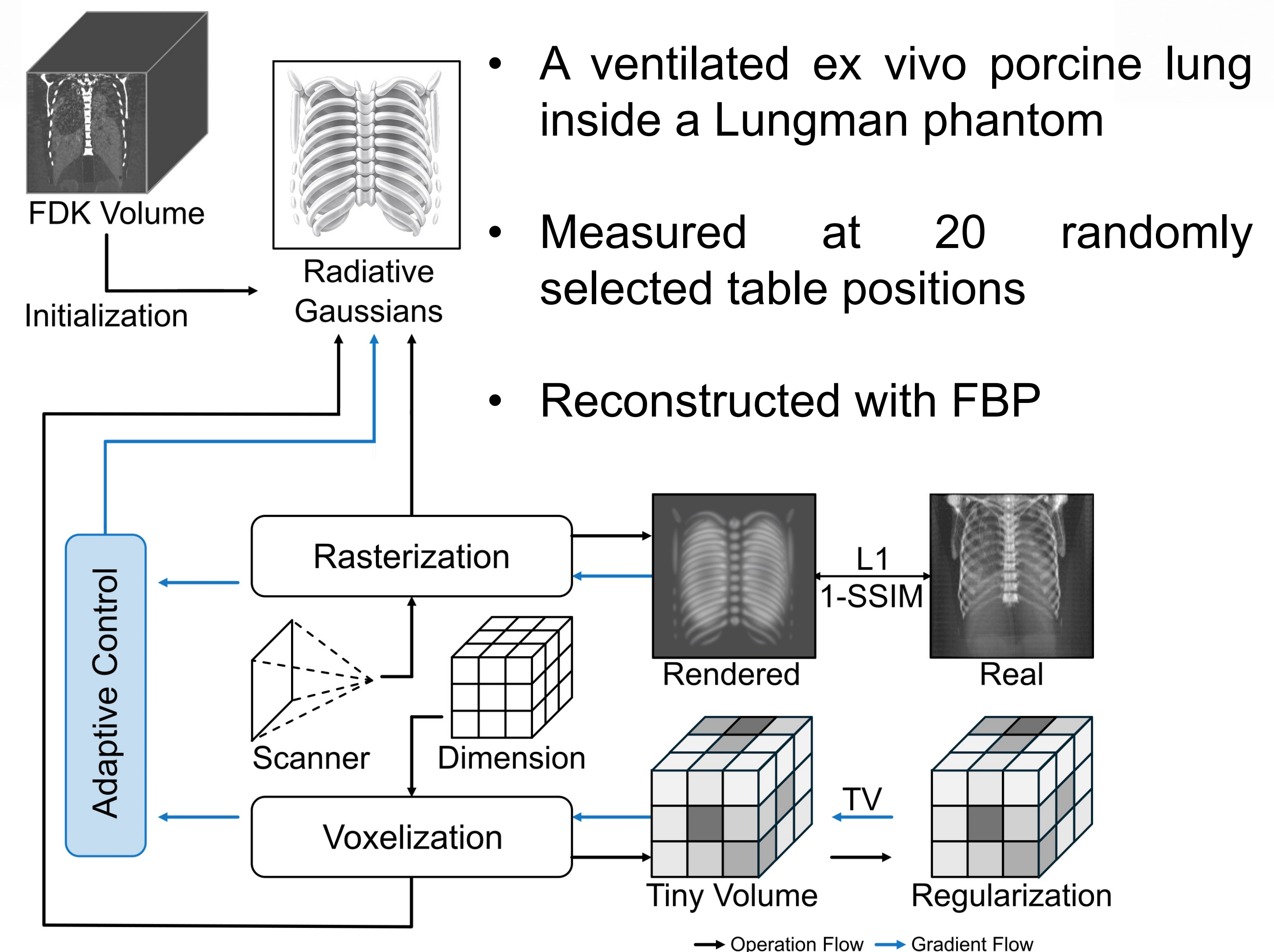
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INTRODUCTION

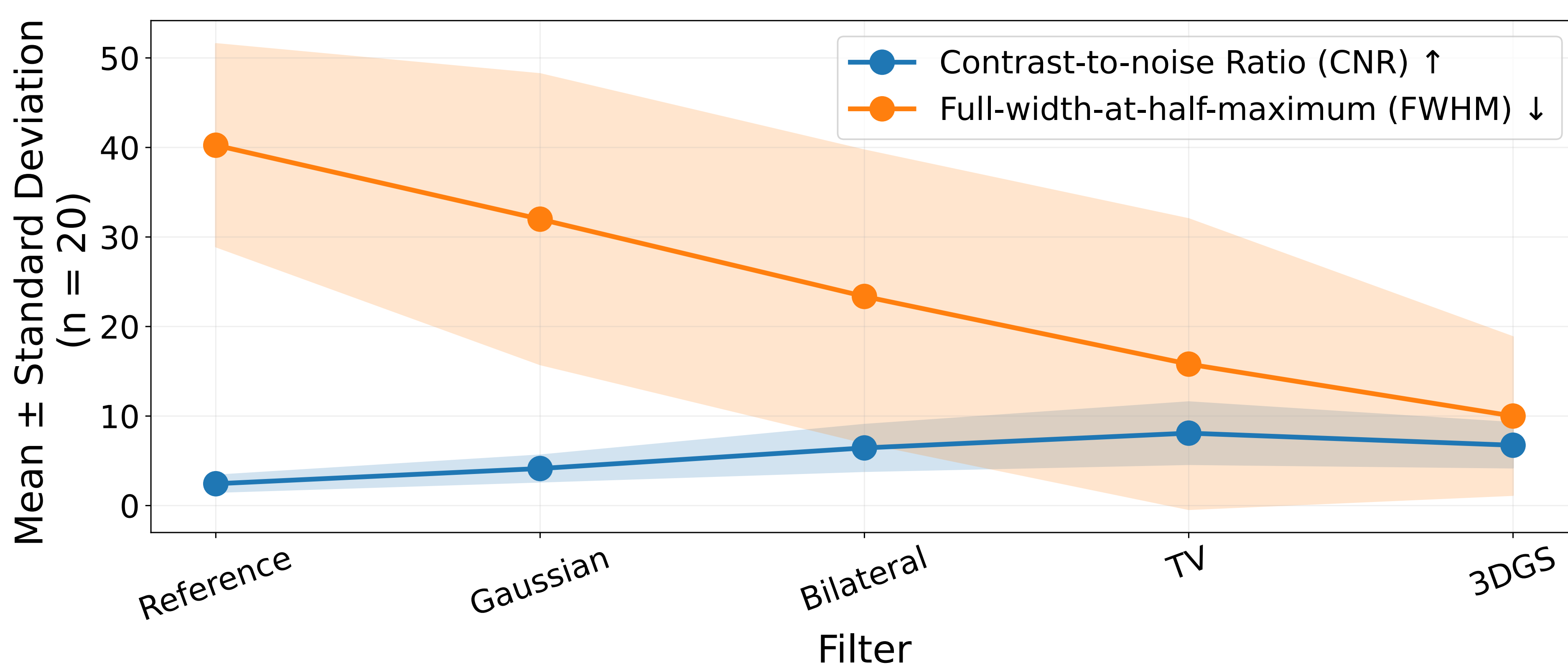
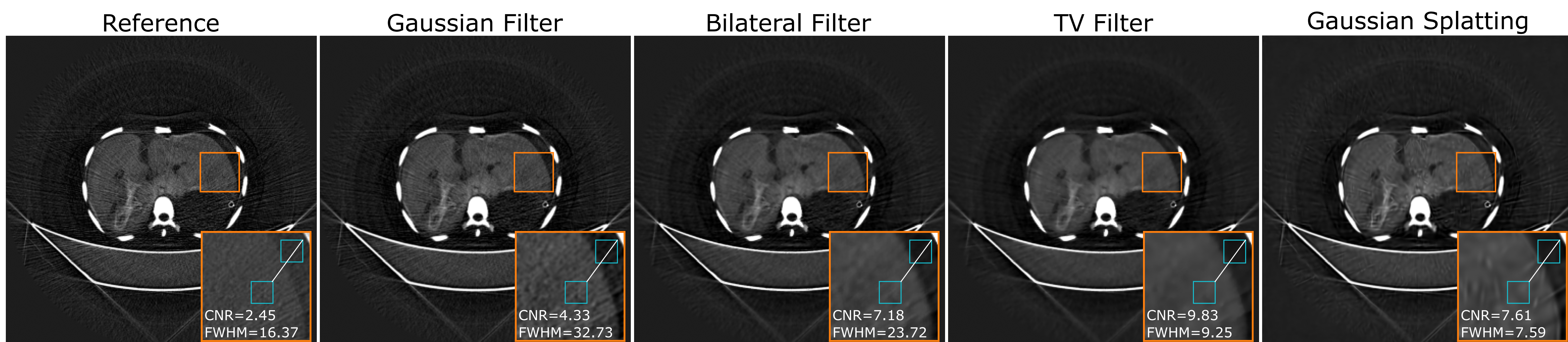
- The clinical dark-field CT scanner (DFCT) enables attenuation- and small-angle scatter-based imaging of human-scale phantoms.
- Due to the scanner's vibrating Talbot-Lau grating interferometer, images suffer from inherent streak artifacts [1].
- In this work, DFCT data is processed by re-purposing the 3D Gaussian Splatting (3DGS) method, designed for rendering sparsely sampled X-ray attenuation data [2], and is compared against conventional filtering:

- Gaussian filter ($\sigma = 1.00$)
- Bilateral filter ($\sigma_{\text{Spatial}} = 5.60$)
- Total Variation (TV) filter ($\lambda = 0.04$)

METHODS



RESULTS



CONCLUSION

These findings indicate that 3DGS outperforms conventional filtering in DFCT streak-reduction, preserving edge sharpness and a high CNR, a necessity for diagnostic accuracy in future clinical trials.

REFERENCES

- [1] M. Viermetz et al., "Dark-field computed tomography reaches the human scale," *PNAS*, vol. 119, 2022.
- [2] R. Zha et al., "R²-gaussian: Rectifying radiative gaussian splatting for tomographic reconstruction," *Advances in Neural Information Processing Systems*, vol. 37, 2024.

Acknowledgements

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